Essentials of Web Programming Lab

Course Outcomes

After completion of course, students would be able to:

1. Design static web pages and forms using HTML elements and controls.
2. Develop styled and responsive web pages using CSS.
3. Implement client-side scripting using JavaScript.
4. Construct XML documents and develop AJAX-based asynchronous web applications.
5. Create single-page web applications using TypeScript and Angular components, forms, and routing.

List of Experiments

Week 1

Develop the static web pages required for an online book store web site: Home Page, Login Page.

Week 2

Create the student Registration Page using HTML forms with basic input validations.

Week 3

Apply internal, external and inline CSS for week1 and week2 pages.

Week 4

Write JavaScript programs to validate the fields of the student registration page.

Week 5

Design the catalogue page for the online book store using HTML, CSS, and JavaScript.

Week 6

Write an XML file which will display the Book information.

Write a Document Type Definition (DTD) to validate the XML file.

Week 7

Review and Skill Test-1

Week 8

Write an Ajax / Fetch API program to load text data asynchronously.

Week 9

Write an Ajax / Fetch API program to parse JSON/XML response data.

Week 10

Write an Ajax / Fetch program to send data using POST method.

Week 11

Develop TypeScript programs using classes, interfaces, and functions.

Week 12-13

Create Angular components and implement data binding, directives.

Week 14

Develop Angular forms, routing, and services for a sample single-page application.

Week 15

Review and Skill Test-2